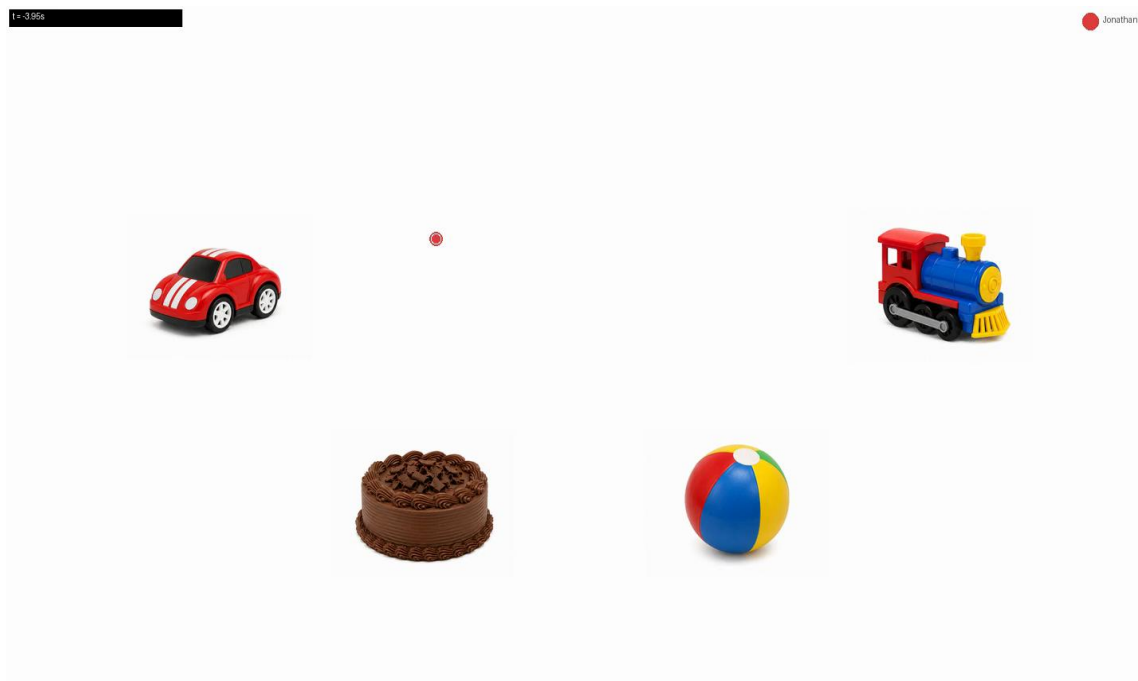


Pilot Data + Preliminary Plots

Acquisition and Analysis of Eye Tracking Data · University of Stuttgart · 2026



Research Question

Are the prediction probabilities of LLMs on subsequent references similarly distributed to probabilities estimated with **human anticipatory eye movements** in a Visual World Paradigm?

2 participants

50 trials

~ 12 mins

sanity check

bugs

Whether the data is actually usable?

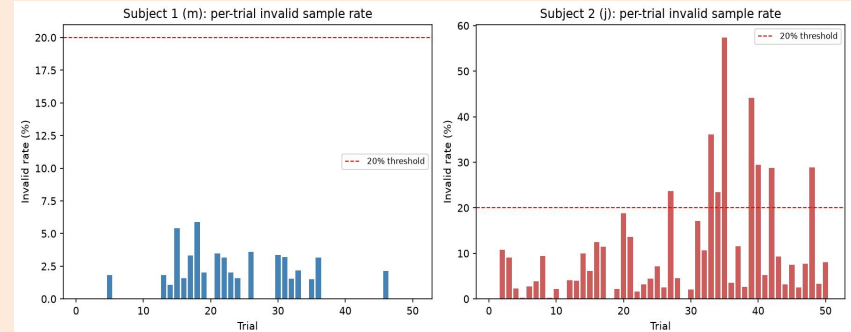
1. Was the tracker set up correctly?

	Left X	Left Y	Right X	Right Y
S1	0.44°	0.60°	0.55°	0.75°
S2	0.49°	0.46°	0.24°	0.78°

Calibration error in degrees (X and Y axes) from pygaze

Both subjects < 1° — within acceptable range

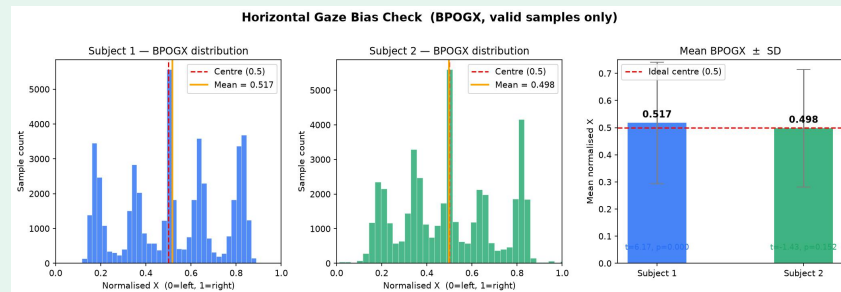
2. Was enough valid data captured?



Per-trial invalid rate (BPOGV=0) — both subjects

S1: 2.3% · S2: 10.5% invalid, maybe break in between

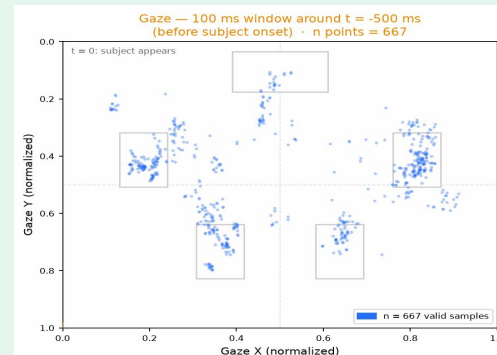
3. Any systematic left/right gaze bias?



BPOGX distribution — mean gaze position ≈ 0.5 for both subjects

no directional bias

4. Does gaze land where the objects actually are?



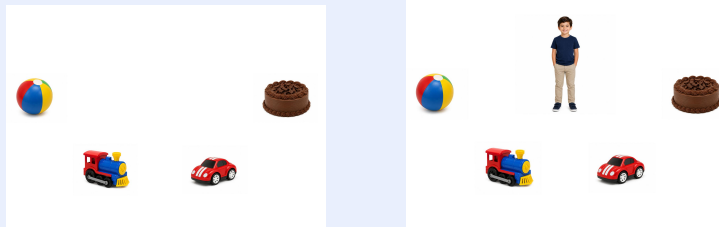
all valid gaze positions across all trials

5 clusters match subject + 4 object positions on screen

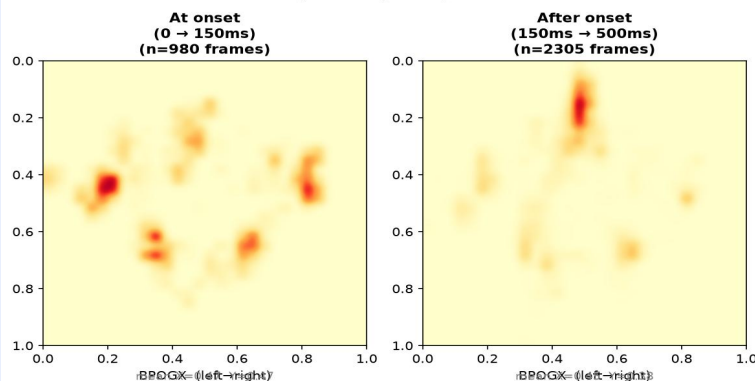
Does the eye movement data actually show what the paradigm is supposed to show?

1. When subject appears on screen

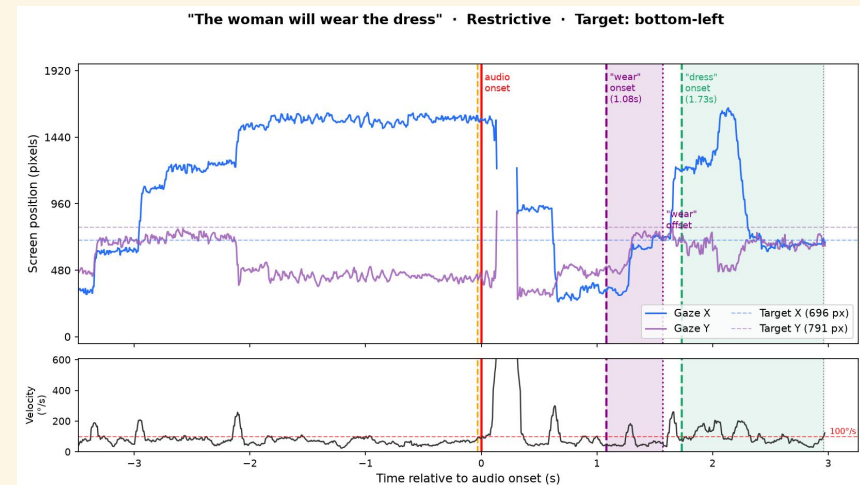
Gaze density shifts from broadly distributed to subject-centered within 150 ms of subject onset . This confirms participants were actively attending to each trial stimulus.



Gaze density — Subject 2, all 50 trials



2. When restrictive verb is heard



Saccades: steep transitions in Gaze X + corresponding velocity confirm rapid, real eye movements

Fixations: flat plateaus between saccades + near-zero velocity confirm stable gaze on a region of interest

Anticipatory gaze: Gaze X begins shifting toward target (696 px) around verb "wear" onset (1.08 s) before the noun "dress" is spoken (1.73 s)

✗ Evidence of data loss - Breaks in the blue line

Logging bug · Stimulus error · Design gap — fixes needed before full data collection

✗ 3 issues identified — gaze stream unaffected, but all must be fixed before full collection

Bug 1 — STOP_TRIAL Trigger Missing in TSV

Expected 50 STOP_TRIAL triggers per subject.

S1: 39/50 present (11 missing)

S2: 28/50 present (22 missing)

Evidence — each row = 1 trial, gaps = missing trigger



S2 (22 missing)

Fix: add some time at the end of trial in OpenSesame inline script

Bug 2 — Stimulus Text Errors

some items in the non-restrictive block have malformed, misleading
eg. id26: "blow grab"
eg. the baker will bake the dough.
eg. subject repeatedly used as object in other sentence

Fix: fix them

Design Gap — No Item Counterbalancing

Both pilot subjects saw identical item–condition assignments: items 1–25 always Restrictive, 26–50 always Non-restrictive.

- Condition is fully confounded with item set.
- Results cannot be generalized across items.

Fix: 2-list Latin square 10 for sentence as well



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Thank you